

L3-4 Judgment Exit Under Continuing Operation

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This set of content operates at the L3 layer. Unlike L1, which establishes methodological frameworks, and L2, which develops judgment models, L3 does not define concepts nor derive mechanisms. It records only the actual behaviors, stage transitions, and resulting outcomes within real projects. What is presented here is not an explanation of the system, but the way in which the system operates under given conditions.

All cases included in this layer are derived from real project processes rather than fictional scenarios. Due to confidentiality requirements, specific details have been appropriately modified, but without altering the actual sequence of events or the structural logic of how they occurred.

In real project environments, certain identified operational requirements or usage conditions do not always enter the system as clearly defined problems. Instead, they are continuously accommodated within existing workflows and carried forward in various forms. Under such conditions, design, coordination, and construction are still able to proceed according to schedule, and the system itself does not become interrupted.

This type of process is not uncommon; however, it is rarely recorded in a complete form.

Within such projects, conditions that have not yet been concretely defined are often handled in interim ways: through spatial reservation, unified assumptions, or by being transferred to later stages of use. These handling methods, once proposed, do not trigger substantive objections and are accepted by all parties involved, thereby entering the design and implementation process. As the project advances, these temporary treatments gradually transform into fixed premises of the system and begin to participate in subsequent work.

During handover and actual operation, some of the conditions that did not enter design-stage judgment reappear within the system, often in the form of spatial adjustments or system modifications. At this point, the system itself does not collapse, but the original premises are no longer sufficient to cover actual usage demands, and the required adjustments must be implemented on top of already completed work.

L3-4 does not document a specific design problem, nor does it evaluate the validity of individual decisions. Instead, it focuses on a persistent condition within project processes: while design, coordination, and implementation continue uninterrupted, certain conditions fail to be resolved at the stage where judgment should occur. These conditions are deferred, substituted, or transferred, and eventually re-enter the system at a later stage.

In the following chapters, this process will be unfolded from different perspectives.

Chapter 1 takes time as the entry point, illustrating how such conditions are carried forward without explicit resolution, and how they re-emerge during actual operation.



Chapter 1

Deferred Judgment: When Unresolved Requirements Are Carried Forward in Time

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Project Background (Factual Summary)

- Project year: 2021
- Location: Southern China (biopharmaceutical industrial park)
- Project type: New-built biopharmaceutical R&D laboratory facility
- Building type: Multi-level laboratory fit-out within an existing completed structure
- Functional composition: Integrated laboratory and administrative (non-lab) spaces
- Project phase: Schematic to detailed design, followed by immediate construction phase
- Current milestone at involvement: Design coordination stage
- Scope of work:
Interface coordination across laboratory planning, MEP systems, and internal functional layout
- Role positioning:
Located at the intersection of laboratory planning and multidisciplinary coordination, responsible for system integration and cross-party alignment (client / consultants / base building design team)
- System characteristics:
Multi-level laboratory system integrated within a single structural framework, with combined experimental workflows, personnel circulation, and logistics paths
- Supporting systems:
HVAC, plumbing, electrical, gas supply, purified water systems, and control systems integrated into a unified operational framework

Chapter 1

Deferred Judgment: When Unresolved Requirements Are Carried Forward in Time

1. Project Entry Under Fixed Schedule and Quantity Constraints

At the point when the project entered the design phase, the overall schedule had already been fixed. Design and construction were structured as a continuous sequence, with construction commencing immediately upon completion of the design phase. No additional time was reserved for major adjustments. Under this condition, all disciplines were required to complete spatial organization, system configuration, and design development within a defined timeframe in order to meet subsequent construction milestones.

At the early stage of design input, the client had already defined key operational requirements for laboratory use, including the handling and storage of reagents. Based on the research functions, experimental types, and personnel scale, the required types and quantities of reagents were estimated and established as part of the operational baseline.

At the same time, reagent storage was required to comply with current regulatory standards governing Class A hazardous materials. Under these constraints, the allowable storage quantity of such reagents is directly linked to the spatial conditions of the laboratory. The total allowable quantity is not an independent variable; it must be determined in relation to the available floor area and corresponding spatial conditions.

As the design progressed, the actual reagent demand was compared against the spatial conditions of the laboratory within the building. This comparison confirmed that, under the current design configuration, the total quantity of reagents required for operation exceeded the allowable limit for storage within the building. At this point, the nature of the problem shifted. The question was no longer whether storage space could be allocated within the laboratory, but rather that, even if such space were provided, it would not be possible to meet the regulatory quantity constraints under the existing building conditions.

This conflict became progressively confirmed during the design process. The project, being a multi-level laboratory facility, had already developed its spatial organization based on experimental functions, equipment layout, and integrated MEP systems. The overall design trajectory had been fixed. Any attempt to resolve the storage issue by reorganizing internal space would require substantial changes to spatial layout, system configuration, and the associated design framework, which did not enter the active design pathway under the fixed schedule.

Through multiple rounds of coordination, a handling direction gradually emerged. Since compliance could not be achieved within the building, the solution shifted toward relocating storage outside the primary structure. The client proposed the construction of a centralized external reagent storage facility at a compliant location outside the building, intended to accommodate the full quantity of required materials. Laboratory users would then retrieve reagents as needed.

The core of this solution did not lie in optimizing internal laboratory organization, but in changing the storage carrier itself, allowing the storage condition to be brought back into regulatory compliance under a different spatial framework.

Following this, the external centralized storage facility was established as a fixed premise for subsequent design and implementation. Internal laboratory layout and system design continued along the original trajectory, and the possibility of in-building storage was no longer pursued. Under the combined conditions of fixed schedule, existing building constraints, and actual reagent demand, a single accepted path was formed: the laboratory would no longer carry the function of centralized storage, and this requirement would instead be transferred to a separately constructed external facility.

2. Acceptance Without Reopening

Once the external centralized storage solution was proposed, discussions quickly shifted toward implementation. After confirming that the solution satisfied regulatory requirements, the client and consultants accepted it as the designated method for addressing reagent storage. In subsequent coordination processes, this solution was no longer treated as one option among alternatives, but was directly adopted as a fixed premise for design progression.

With this premise established, the focus of design discussions shifted accordingly. The earlier question of whether storage could be resolved within the building gradually exited the scope of coordination.

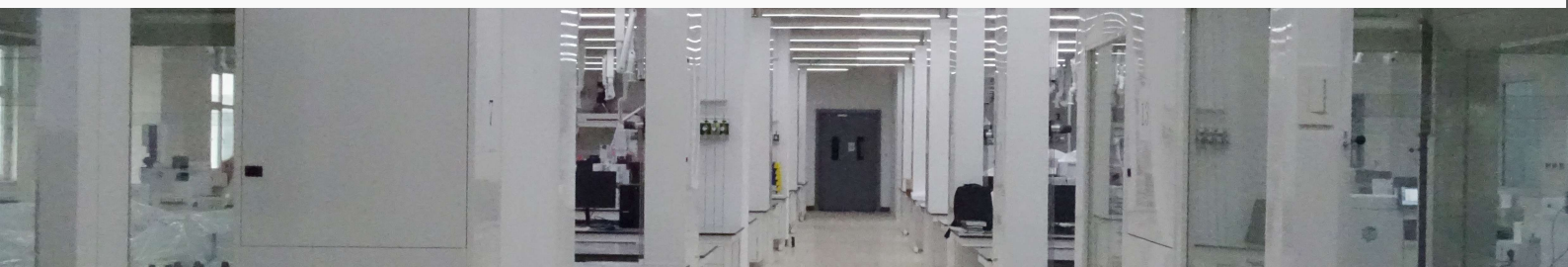
Conversations no longer returned to the possibility of internal spatial allocation, but instead concentrated on the configuration of the external facility itself. Topics such as site selection, storage capacity, classification management, and operational procedures became the primary subjects of discussion, all framed within the requirement to deliver the solution within the existing schedule.

During this process, the internal laboratory layout and system design continued along their original trajectory. As the storage issue had been transferred externally, no adjustments were made to internal spatial configuration in response to it. Design work was not interrupted or rolled back due to this issue. Instead, coordination efforts and design resources increasingly aligned with the accepted solution, further reinforcing its position within the project.

At the same time, based on prior laboratory project experience, an alternative approach was proposed from the design side: the introduction of reagent buffer rooms within the laboratory. This strategy was based on limiting storage to a 24-hour operational quantity within each laboratory area, thereby avoiding the total quantity threshold imposed by regulations through distributed, time-based control. Under this approach, compliant storage could potentially be achieved within the building.

Although this alternative entered discussion, it was not incorporated into the subsequent design pathway. The proposal did not trigger a re-evaluation of internal storage feasibility, nor did it lead to any adjustment of the established solution. The external centralized storage remained as the governing premise, while the buffer-room approach did not proceed into systematic analysis or design development, and gradually faded out of coordination discussions.

Following this, both design and implementation continued to advance along the accepted path. The external storage facility and the laboratory building progressed in parallel into detailed design and construction preparation, with no further divergence or reopening of the storage strategy.



3. Design Fixation Under an Accepted Premise

After the external centralized storage solution was confirmed and entered the design process, the corresponding handling approach gradually stabilized as a governing premise within the project. As design work continued to develop, the laboratory's primary spatial organization and all associated engineering systems moved into the detailed design phase, where spatial arrangements and system configurations were progressively refined toward construction readiness.

At this stage, the focus of design work shifted from selecting between possible approaches to executing an already accepted path. Internal laboratory zoning, equipment layout, and the integration of HVAC, electrical, plumbing, and other supporting systems were all carried forward based on the established assumption that the building would no longer accommodate centralized reagent storage. Drawings and system parameters were developed accordingly and moved toward finalization. In parallel, the external storage facility was developed as a fixed component of the overall system, with its location, capacity, and operational configuration progressively defined and prepared for implementation.

As multidisciplinary coordination advanced, the accepted path became increasingly embedded across multiple layers of the project. On one hand, internal laboratory spaces were organized under the premise that no centralized storage function would be accommodated within the building, and no corresponding spatial reservations were maintained. On the other hand, the external facility became structurally integrated into the operational logic of the project, forming a separate storage carrier that functioned independently from the laboratory interior. In this process, reagent storage was no longer treated as a design variable subject to adjustment, but rather as an established system condition upon which other decisions were based.

During coordination and documentation, discussions related to alternative storage approaches became noticeably reduced. Communication among parties focused primarily on detailing and aligning the accepted solution, including operational workflows between the laboratory and the external facility, transportation routes, and management procedures. Topics that could have introduced alternative paths did not re-enter the coordination scope, nor were provisions left in drawings or system configurations that would allow for future reconsideration within the current design stage.

As design documentation was completed and transitioned into construction, the established path was further solidified. Construction preparation and on-site implementation were executed based on finalized design outputs, with the laboratory interior constructed according to the defined layout and system configuration, and the external storage facility built in parallel. At this point, the storage strategy had reached closure across both design and construction phases, and no conditions remained under which the path could be reopened within the ongoing project timeline.

Under this state, reagent storage ceased to exist as an issue requiring active judgment. Instead, it became a fixed operational premise embedded within the system, participating in subsequent implementation without further evaluation or reconsideration.

4. Post-Occupancy Re-entry and System Reversal

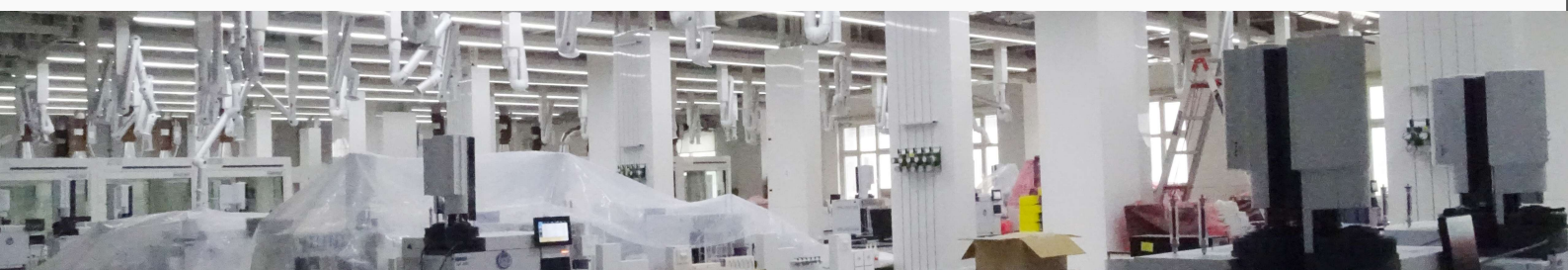
Following project completion and handover, the laboratory entered operation under the predefined design framework. The external centralized storage facility was put into use in parallel with the main building, and reagent handling followed the established retrieval process. Internal laboratory spaces and systems remained unchanged during the initial stage, with storage fully dependent on the external facility. Over approximately seven months of operation, a persistent misalignment emerged between reagent access and experimental workflow. Reagent retrieval functioned as a separate process, introducing repeated interruptions, extended circulation paths, and increasing time and labor costs. These effects accumulated gradually, as daily operations continued to favor experimental continuity over adherence to the predefined storage pathway.

Although the issue was raised repeatedly, no immediate adjustment was made. The system continued to operate under the established premise until the mismatch between storage configuration and actual use became collectively recognized, ultimately triggering a structural revision.

At this stage, a previously unadopted approach was reintroduced: the use of internal reagent buffer rooms, each designed to accommodate approximately one day of usage. This allowed storage to re-enter the laboratory in a distributed and time-limited form, restoring proximity between storage and use while maintaining compliance with overall quantity constraints.

The project then entered a partial reconstruction phase. Existing laboratory spaces were selectively demolished and reorganized, and internal circulation was adjusted accordingly. On each floor, two independent buffer rooms were installed at opposite ends of the floor plate, forming a distributed storage structure capable of serving different laboratory zones. This configuration confined reagent access within the same level, replacing cross-building movement with localized circulation.

Following these modifications, the storage system evolved into a dual-layer structure: the external facility retained responsibility for total volume control, while internal buffer rooms supported high-frequency usage. Operational continuity was restored, and the system regained alignment with actual working conditions. However, as these changes were implemented after the system had already been fully designed and constructed, the resulting configuration remained constrained by existing spatial and infrastructural conditions. Buffer rooms were inserted into an already fixed layout, and circulation could only be adjusted within predefined pathways. The outcome is therefore not an optimized solution, but a reconstructed condition that restores functional viability within existing constraints.



Chapter 1 — Conclusion

In this case, the issue of reagent storage was not resolved through spatial reorganization during the design phase, but was instead displaced by redefining the storage carrier. Under fixed schedule constraints and regulatory limits, this displacement was accepted as a workable solution and progressively stabilized as a governing premise within the system.

As the project advanced, this premise became embedded across spatial, technical, and operational layers. Internal laboratory space was organized without accommodating centralized storage, while the external facility assumed full responsibility for compliance. Alternative approaches were proposed but did not trigger re-evaluation, and no structural interface was preserved for their later inclusion. As a result, the system proceeded toward completion along a single accepted path, without further judgment.

During operation, this path was continuously carried forward and tested under real usage conditions. The separation between storage and use introduced persistent inefficiencies, which accumulated over time and gradually manifested as systemic pressure. Although these conditions were recognized during operation, they did not immediately lead to revision, allowing the system to continue functioning under its established premise.

Only after sustained operational misalignment did the previously excluded condition re-enter the system. Through partial reconstruction and the insertion of distributed storage nodes, the system regained functional alignment with actual use. However, this adjustment occurred within an already fixed spatial and infrastructural framework, resulting in a recovered configuration rather than an optimized one.

In this sense, the process observed in this case is not defined by a single decision, but by a sequence of transformations: a condition exits judgment, is carried forward as an assumption, accumulates as operational cost, and eventually returns as a structural requirement.

When critical conditions are deferred as “non-immediate” during design, they are not removed from the system; they persist as latent constraints and ultimately return through operational cost, requiring resolution at a higher structural **expense**.

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